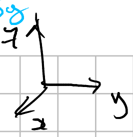
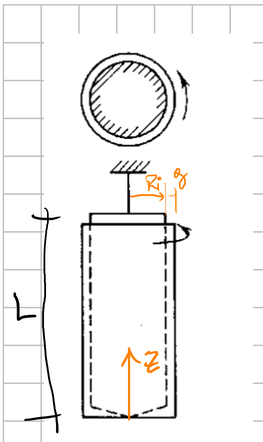
$\tau = \mu \cdot \frac{\partial v_x}{\partial y}$



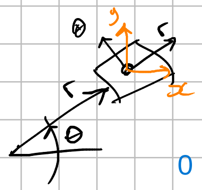


Ec. de Stokes para fluidos newtonianos

$$\tau_{ij} = -p\,\delta_{ij} + 2\mu V_{ij} - \frac{2}{3}\mu V_{kk}\,\delta_{ij}$$

$\mu : V$

el ser isotropico

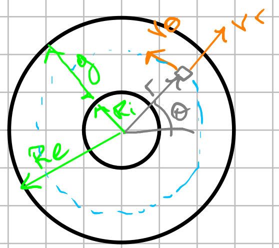


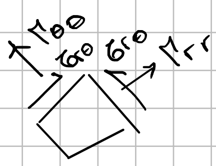
0 por delta kronecker

$$\tau_{r\theta} = -p\,\overset{0}{\delta_{r\theta}} + 2\mu\, V_{r\theta} - \frac{2}{3}\mu\left(V_{rr}+V_{\theta\theta}+V_{zz}\right)\cdot \overset{0}{\cancel{\delta_{r\theta}}}$$

$$\tau_{r\theta} = 2\mu V_{r\theta}$$

$$V_{r\theta} = \frac{1}{2}\left(\frac{1}{r}\frac{\partial v_r}{\partial \theta} + \frac{\partial v_\theta}{\partial r} - \frac{v_\theta}{r}\right)$$





vr de la velocidad
a medida que cambiamos el tita vr no cambia por simetria axial

$$\frac{\partial v_r}{\partial \theta} = 0$$ Por simetrie axial

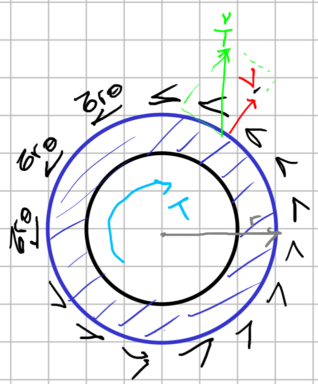
$$R_i >> g \Rightarrow \frac{v_\theta}{r} \simeq 0$$

$$V_{r\theta} = \frac{1}{2}\frac{\partial v_\theta}{\partial r}$$

Ec. de Newton para fluidos

$\tau = \mu \cdot \frac{dv_x}{dy}$

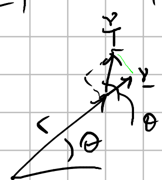
$$\tau_{r\theta} = \cancel{2}\mu \cdot \frac{1}{\cancel{2}} \cdot \frac{\partial v_\theta}{\partial r} = \mu \cdot \frac{\partial v_\theta}{\partial r}$$



$$-\oint_S \underline{r} \times \underset{\sim}{T}\, ds + T = 0$$

$$|\underline{r} \times \underset{\sim}{T}|\, ds = |\underline{r}| \cdot \underbrace{|\underset{\sim}{T}|\, \text{sen}\,\theta}_{\tau_{r\theta}} \cdot ds$$ (dF)

↳ componentes tangencial



integramos la superficie

$$T = \int_{z=0}^{z=L}\int_{\theta=0}^{\theta=2\pi} r \cdot dF = \int_0^L \int_0^{2\pi} r \cdot \tau_{r\theta} \cdot \underbrace{r\,d\theta \cdot dz}_{dS} =$$

$$= \tau_{r\theta} \cdot r^2\, 2\pi L \Rightarrow \tau_{r\theta} = \frac{T}{r^2\, 2\pi L}$$

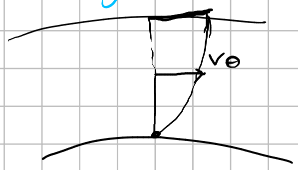
$$\frac{\partial v_\theta}{\partial r} = \frac{T}{2\pi r^2 L \mu}$$

$$\int_{R_i}^{r} \frac{\partial v_\theta}{\partial r} dr = V_\theta(r) - V_\theta(R_i) = \int_{R_i}^{r} \frac{T}{2\pi r^2 L\mu} dr = \frac{T}{2\pi L\mu}\left(\frac{1}{R_i} - \frac{1}{r}\right)$$

Condición de borde $\Rightarrow V_\theta(R_i) = 0$

(por la condición de no deslizamiento y el cilindro interno está fijo)

no es lineal por 1/r



Perfil de velocidad $\quad V_\theta(r) = \frac{T}{2\pi L\mu}\left(\frac{1}{R_i} - \frac{1}{r}\right)$

conocido

$$V_\theta(R_e) = \omega . R_e = \frac{T}{2\pi L\mu}\left(\frac{1}{R_i} - \frac{1}{R_e}\right)$$

Viscosidad del fluido $\rightarrow \mu = \frac{T}{2\pi L\omega R_e}\left(\frac{1}{R_i} - \frac{1}{R_e}\right) \longrightarrow$ Ensayo

$$T = \mu \cdot \frac{2\pi L\omega R_e}{\left(\frac{1}{R_i} - \frac{1}{R_e}\right)}$$

$$T = f(\omega)$$